

Algebra II with Trigonometry

Chapter 6.1

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:

[4506904_alg-2trig-h-chp-6-1-note-docx.pdf](#)

Problems

1. Suppose the terminal point $P(x, y)$ determined by a real number t lies in the first quadrant and satisfies $x + y = \frac{7}{5}$. Evaluate $x - y$ given that $x > y$.
2. Let $P(x, y)$ be a terminal point on the unit circle. Determine all possible values for the reference number $\bar{t}$ associated with P if its coordinates satisfy the equation $x^3 + y^3 = 1$.
3. Let x_k be the x -coordinate of the terminal point determined by the real number $t_k = \frac{k\pi}{8}$. Mentally evaluate the product $x_1x_2x_3x_5x_6x_7$.
4. A terminal point P on the unit circle is determined by the real number $t = 11$. Using estimation, determine exactly which quadrant contains P .
5. Let $P_1(x_1, y_1)$ be the terminal point determined by the real number $t_1 = t$. A sequence of terminal points is generated such that P_{n+1} is determined by $t_{n+1} = \frac{\pi}{2} - t_n$ if n is odd, and $t_{n+1} = \pi + t_n$ if n is even. If $P_1 = (\frac{3}{5}, \frac{4}{5})$, determine the exact coordinates of P_{2024} .
6. Find the sum of the reference numbers $\bar{t}$ for all terminal points $P(x, y)$ on the unit circle whose x -coordinates satisfy the equation $16x^4 - 16x^2 + 3 = 0$.